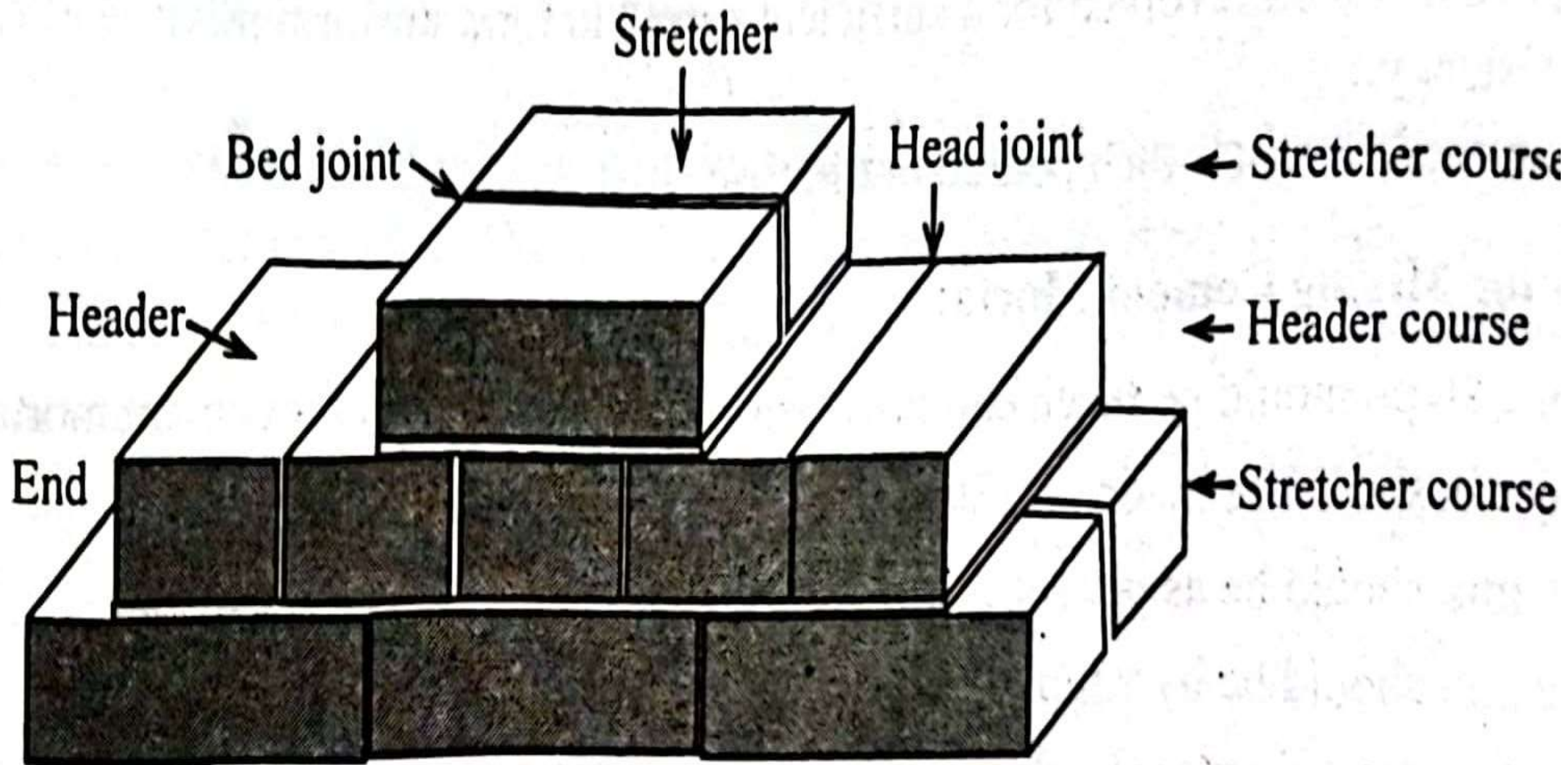
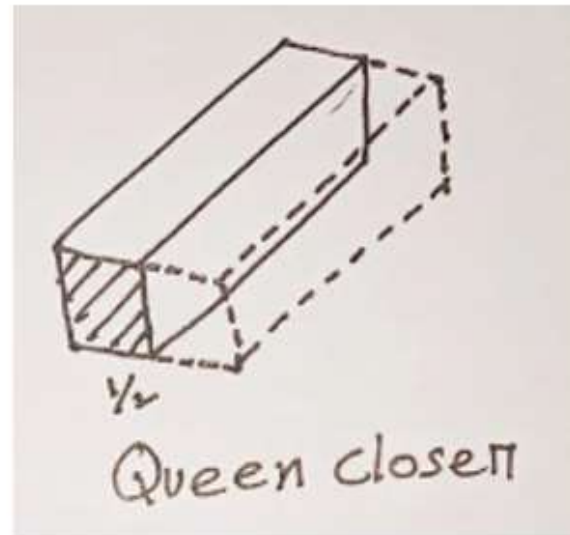
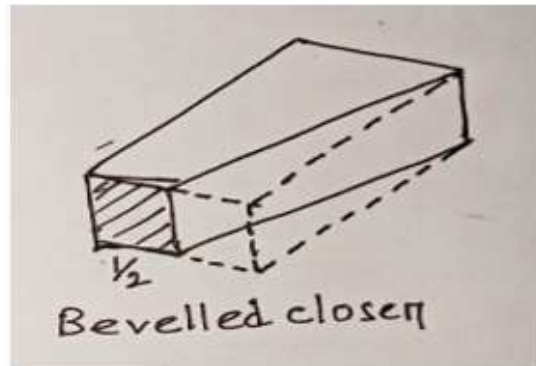
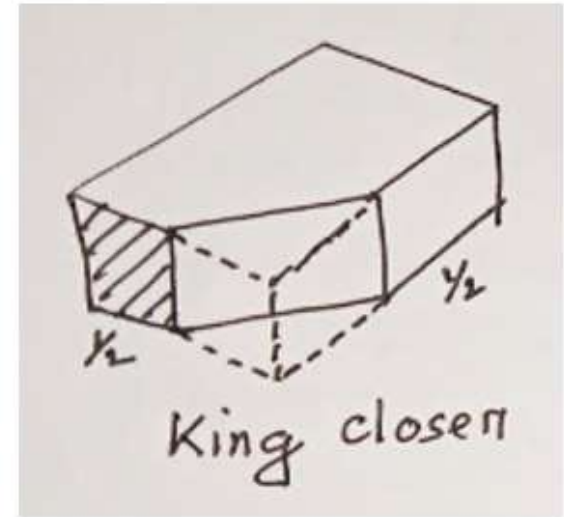
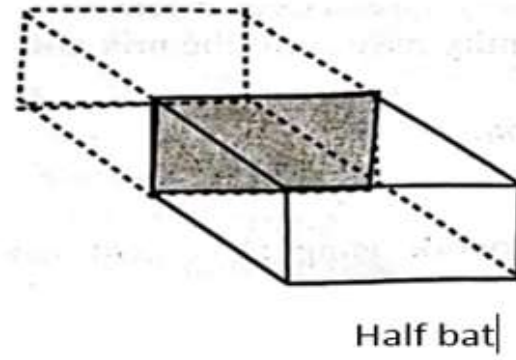
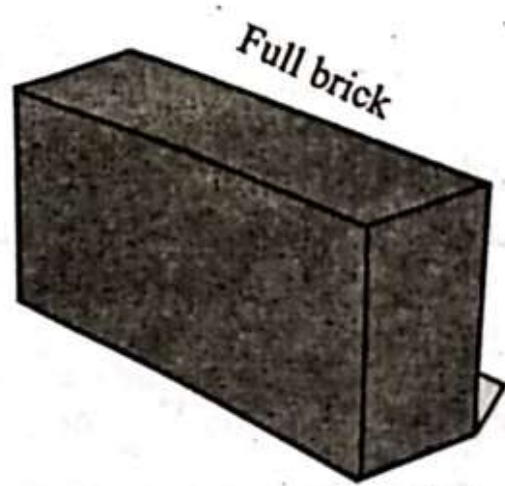


Superstructure works

Brick Masonry:

- Brick units bonded together with mortar to form different structure during construction is termed as brick masonry.
- Brick masonry strength depends on the quality of the brick used, types of mortar and method of bonding used in the construction.





Brick Bonding

- Bonding is an arrangement of bricks in a wall, column or pier laid to a set pattern to maintain adequate lap.

Purposes of Brick Bonding

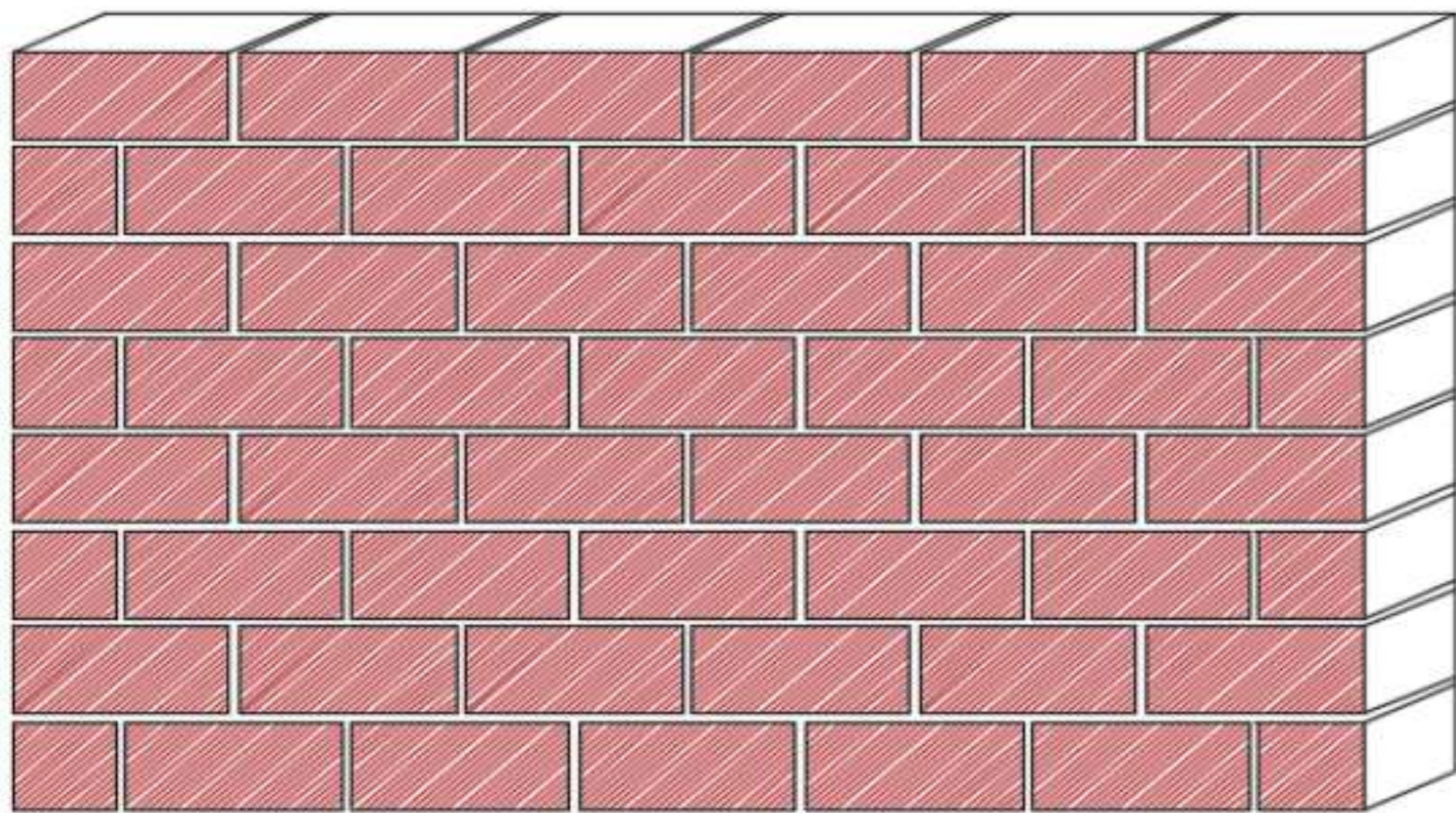
- Obtain maximum strength for distributing the loads to be carried throughout the wall column or pier.
- Ensure lateral stability and resistance to side thrusts.
- Create an acceptable appearance.

Types of Bond

(a) Stretcher Bond

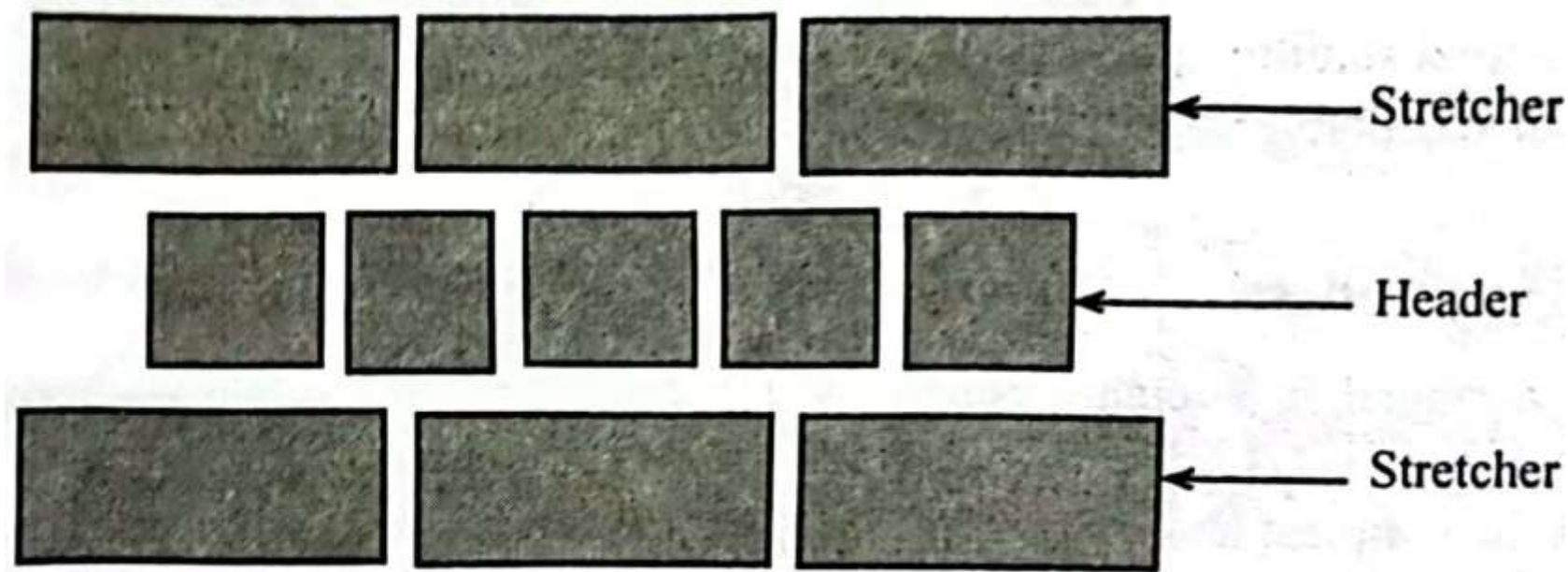
All bricks are arranged in stretcher course in this bond. The following are some features of stretcher bond:

- This is the simplest and most widely used bond.
- All bricks laid in stretcher course create stretcher bond.
- This bond is useful for half brick wall, partition walls, cavity wall, etc.
- There is no header in such walls.



(b)English Bond

- Consists of headers and stretchers laid in alternative courses.
- Strongest of all bonds.
- Much attention is not required in providing this bond.
- Progress of work is faster.



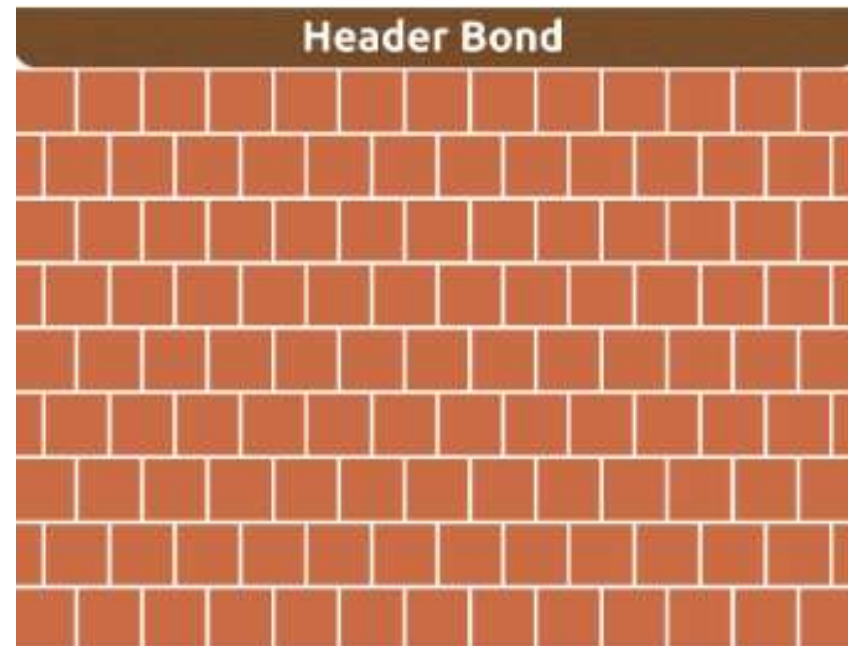
(c)Flemish Bond

- Consists of headers and stretchers laid alternative in courses.
- Provides good appearance especially for all thickness of walls.
- Progress of work is slow.
- Economical because use of brick bats is allowed for forming this bond.



(d)Header Bond

- A header bond in brick masonry is a pattern where all bricks in every course are laid with their shorter face (the "header") exposed on the face of the wall.
- It is primarily used for constructing one-brick-thick walls, footings, and curved structures.



Specification for Brick Masonry Work

The brick masonry shall be built with the usually specified care regarding pre-soaking of bricks in water, level bedding of planes fully covered with mortar, vertical joints broken from courses to courses and their filling with mortar fully.

Bricks:

- Bricks shall be of a standard rectangular shape, burnt red, hand formed or machine made and of crushing strength not less than 3.5N/mm^2 .
- The higher the density and strength the better they will be.
- According to National Building Code (NBC 205 : 1994) The standard size of brick in Nepal is $240 \times 115 \times 57$ mm with 10 mm thick horizontal and vertical mortar joints. Tolerances of -10 mm on length, -5 mm on width.

First Class Brick Work

- It is made by using first class bricks and cement mortar.
- It is made in rich mortar in which the cement and sand ratio is from 1:3 to 1:6.

The characteristics of first class brick are as follows:

- **Color:** Should be well burnt having uniform red color.
- **Size and Shape:** Should be of same standard size, have smooth surface & sharp edges, free from cracks.
- **Soundness:** When struck together, should give metallic sound.
- **Hardness:** Should be hard that it could not be scratched by fingernails.
- **Porosity:** It should not absorb water more than 15% of its dry weight when immersed in water for 24 hrs.
- **Strength:** Minimum crossing strength should not be less than 105 kg/cm^2 (10 N/mm^2).
- **Durability:** Should be able to resist the effects of weathering agencies
- **Efflorescence:** Should not contain much alkaline salts which may cause efflorescence on its surface and decay the bricks.

Second Class Brick Work

- Second class brick work is made by using second class bricks and cement mortar.
- These bricks also have the property of first class bricks but are not very regular or even in shape.
- It should not absorb water more than 20% of its dry weight when immersed in water for 24 hrs.
- Second class bricks have minimum crushing strength 70kg/cm^2 (7N/mm^2).

Third Class Brick Work

- This type of brick work is made by using third class bricks and cement mortar or mud mortar.
- Generally, this type of brick work is made for temporary work.
- These bricks have minimum crushing strength 35 kg/cm^2 (3.5 N/mm^2).

Stone Masonry:

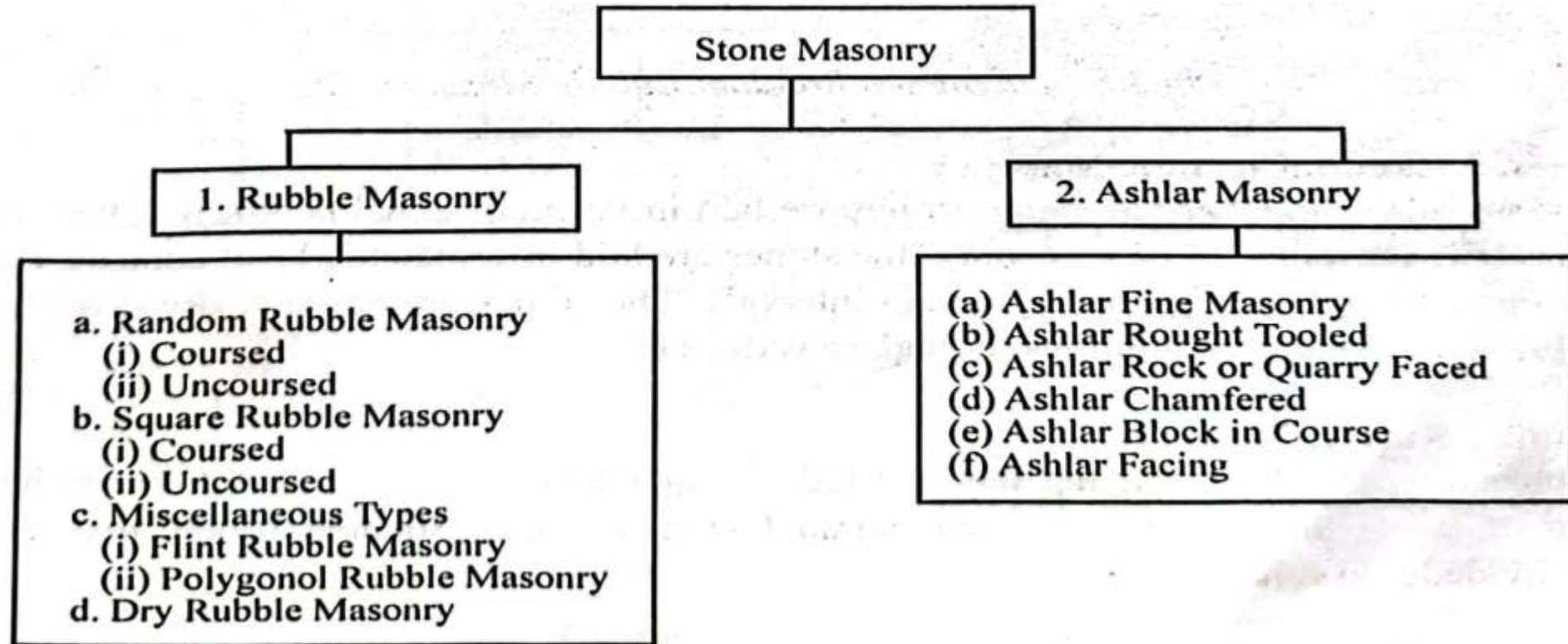
- Stones used as the building units are known as stone masonry.
- The stones used for masonry should be hard, durable, tough, sound and free from weathering decay.

Advantages of stone masonry structures:

- Helps to maintain thermal comfort in the building.
- Reduces cost of painting.
- Good fire protection and heat resistance.
- Long lifespan.
- Attractive appearance.
- In the context of Nepal locally/easily available.

Types of Stone Masonry:

- Depending upon the arrangement of stones in the construction, degree of refinement used in shaping and finishing, stone masonry can be classified as follows:



1. Rubble Masonry

- The stone masonry in which either undressed or roughly dressed stone are laid in a suitable mortar is called rubble masonry.
- This is the roughest and cheapest form of stone masonry.
- In this masonry the joints do not have uniform thickness.

It is divided into the following types:

(a) Random Rubble Masonry

- The rubble masonry in which either undressed or hammer dressed stones are used is called random rubble masonry.

Further random rubble masonry is also divided into the following types:

(i) Uncoursed Random Rubble Masonry

- The random rubble masonry in which stones are laid without forming courses.
- This is the roughest and cheapest type of masonry.
- The stones used in this masonry are of different sizes & shapes.
- Large stones are used at corners to increase their strength.
- It is used for construction of walls of low height in case of ordinary buildings.



1. Coursed Rubble Masonry



3. Polygon Rubble Masonry

What is Rubble Masonry & Its Types



2. Uncoursed Rubble Masonry



3. Polygon Rubble Masonry

(ii) Coursed Random Rubble Masonry

- The random rubble masonry in which stones are laid in layers of equal.
- In this masonry, the stones are laid in somewhat level courses.
- The stones are hammer dressed. Used for construction of residential buildings, boundary walls etc.

(b) Squared Rubble Masonry

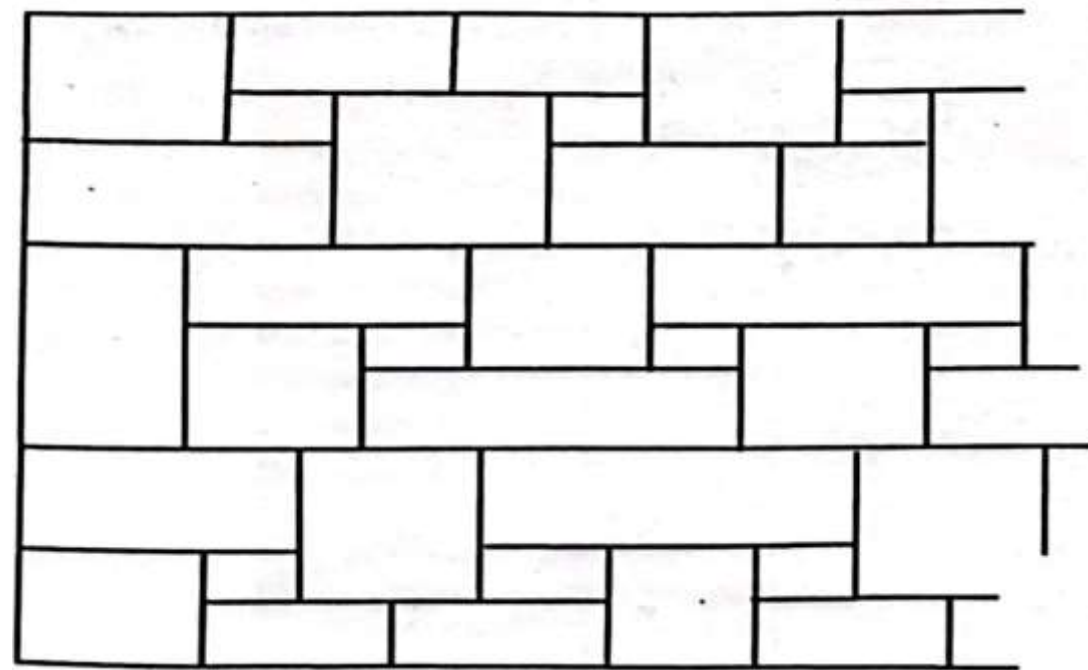
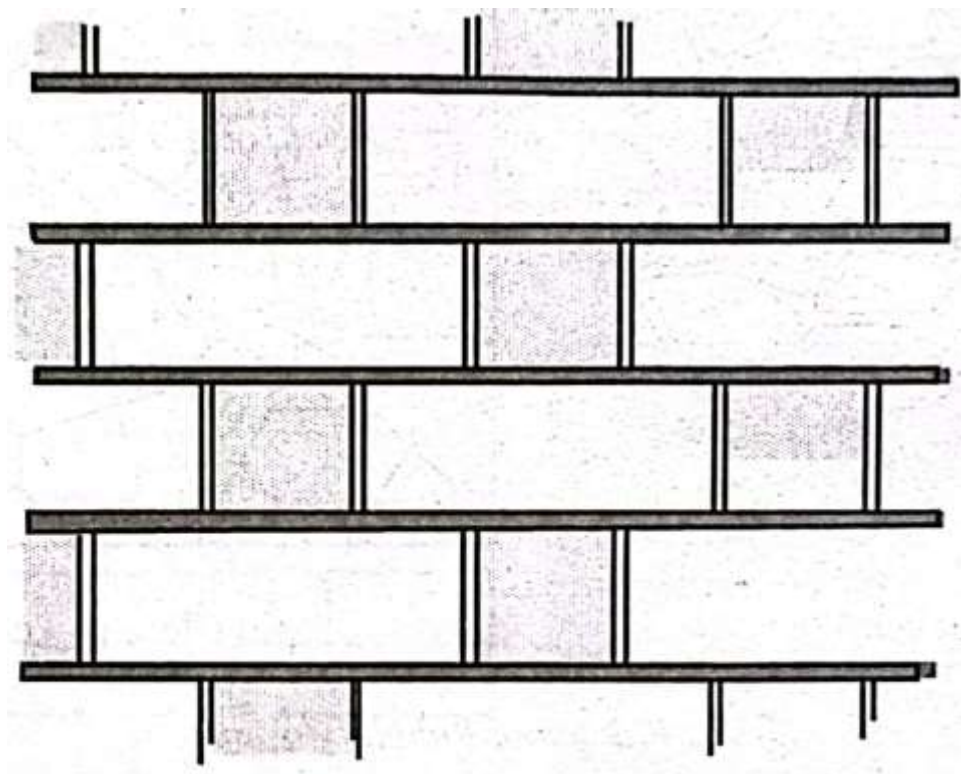
- The rubble masonry in which the face stones are squared on all joints and beds by hammer dressing or chisel dressing before their actual laying.
- It is further divided into following types:

(i) Coursed Square Rubble Masonry

- In which chisel dressed stones laid in courses is called coarse square rubble masonry.
- This is a superior variety of rubble masonry.
- It consists of stones, which are squared on all joints and laid in courses.
- Used for construction of public buildings, hospitals, school, markets, modern residential buildings etc.

(ii) Uncoursed Square Rubble Masonry

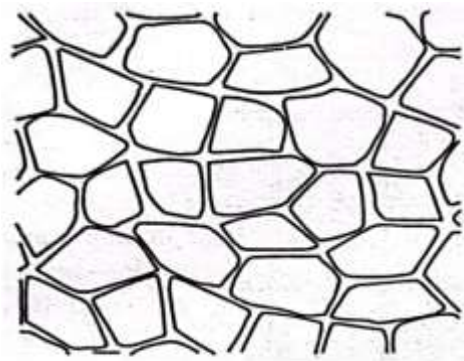
- In this type of masonry hammer dressed stones are laid without making courses.
- It consists of stones which are squared on all joints and beds by hammer dressing.
- All the stones to be laid are of different sizes.
- Used for construction of ordinary buildings in hilly areas where a good variety of stones are cheaply available.



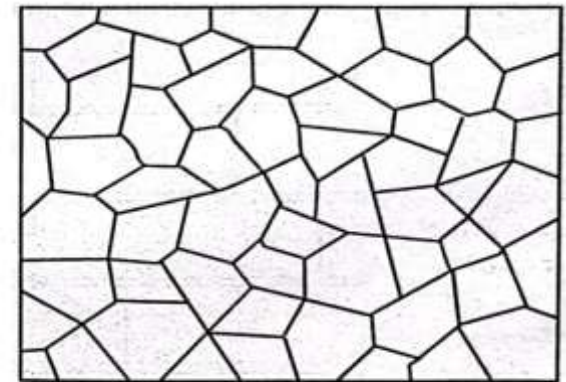
(c) Miscellaneous Types:

(i) Polygonal Rubble Masonry:

- In this type of masonry stones are hammer finished on face to an irregular polygonal shape.
- Stones are bedded in position to show face joints running irregularly in all directions.
- Two type of polygonal walls as rough picked and close picked may be constructed.



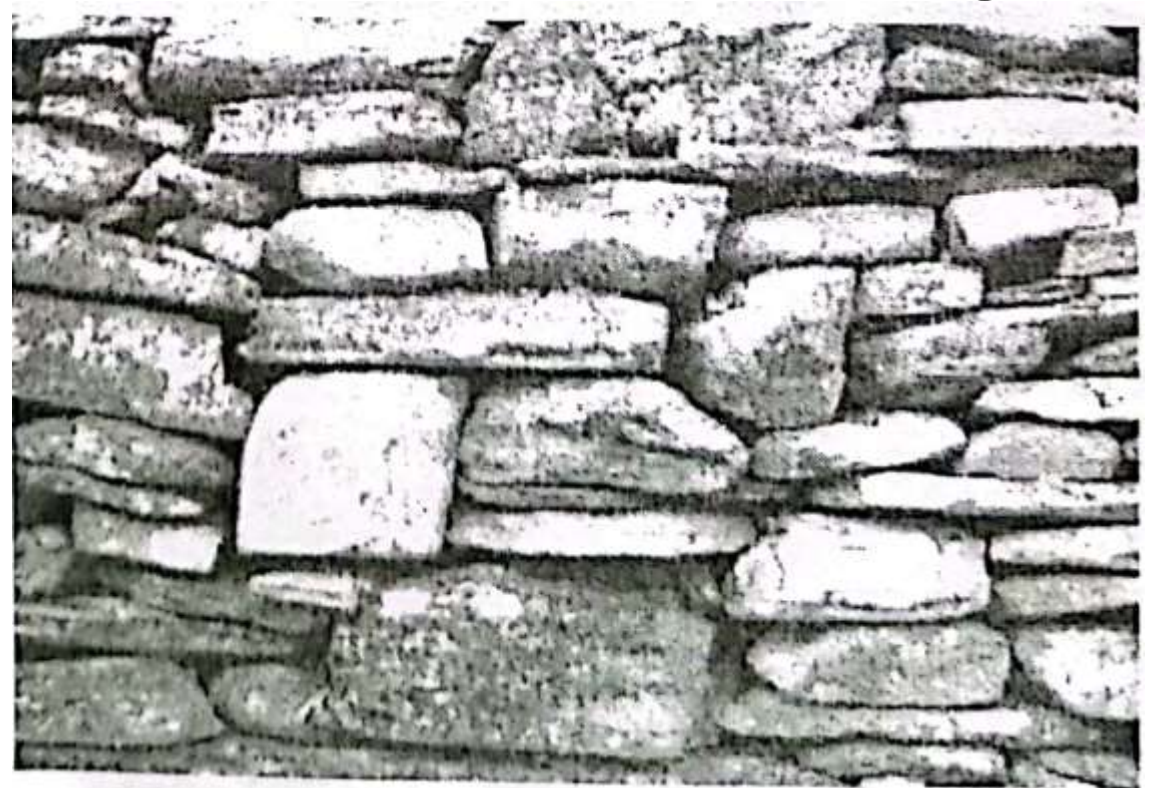
(a) Rough picked



(b) Close picked

(d) Dry Rubble Masonry

- In which stones are laid without using any mortar is called dry rubble masonry or sometimes shortly as "dry stones".
- It is an ordinary masonry and is recommended for constructing walls of height not more than 6m.



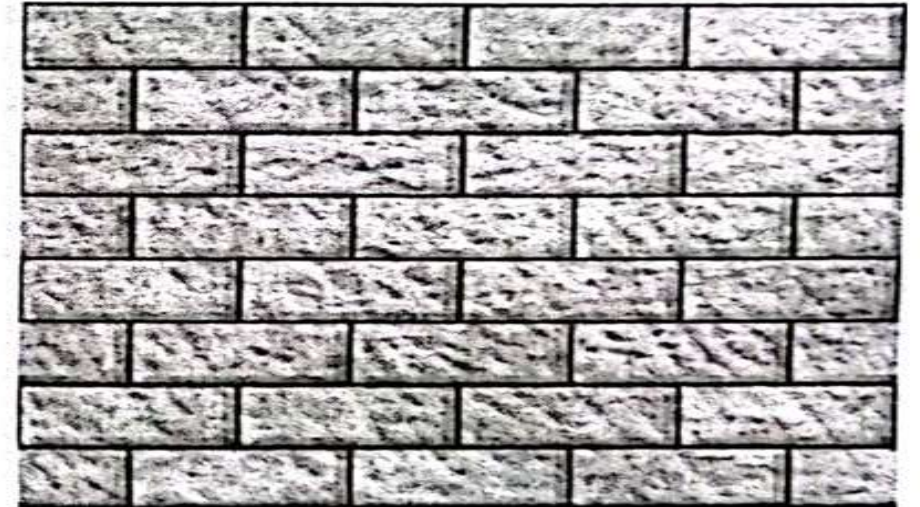
2.Ashlar Masonry

- In which finely dressed stones are laid in cement or lime mortar is known as ashlar masonry.
- In this masonry the courses are of uniform height, all the joints are regular, thin and have uniform thickness.
- This type of masonry is much costly as it requires dressing of stones.
- This masonry is used for heavy structures, architectural buildings, high piers and abutments of bridges.
- Ashlar masonry is classified into following types:

(a) Ashlar Fine (Regular/Random coursed)

(i) Regular Coursed Ashlar Masonry

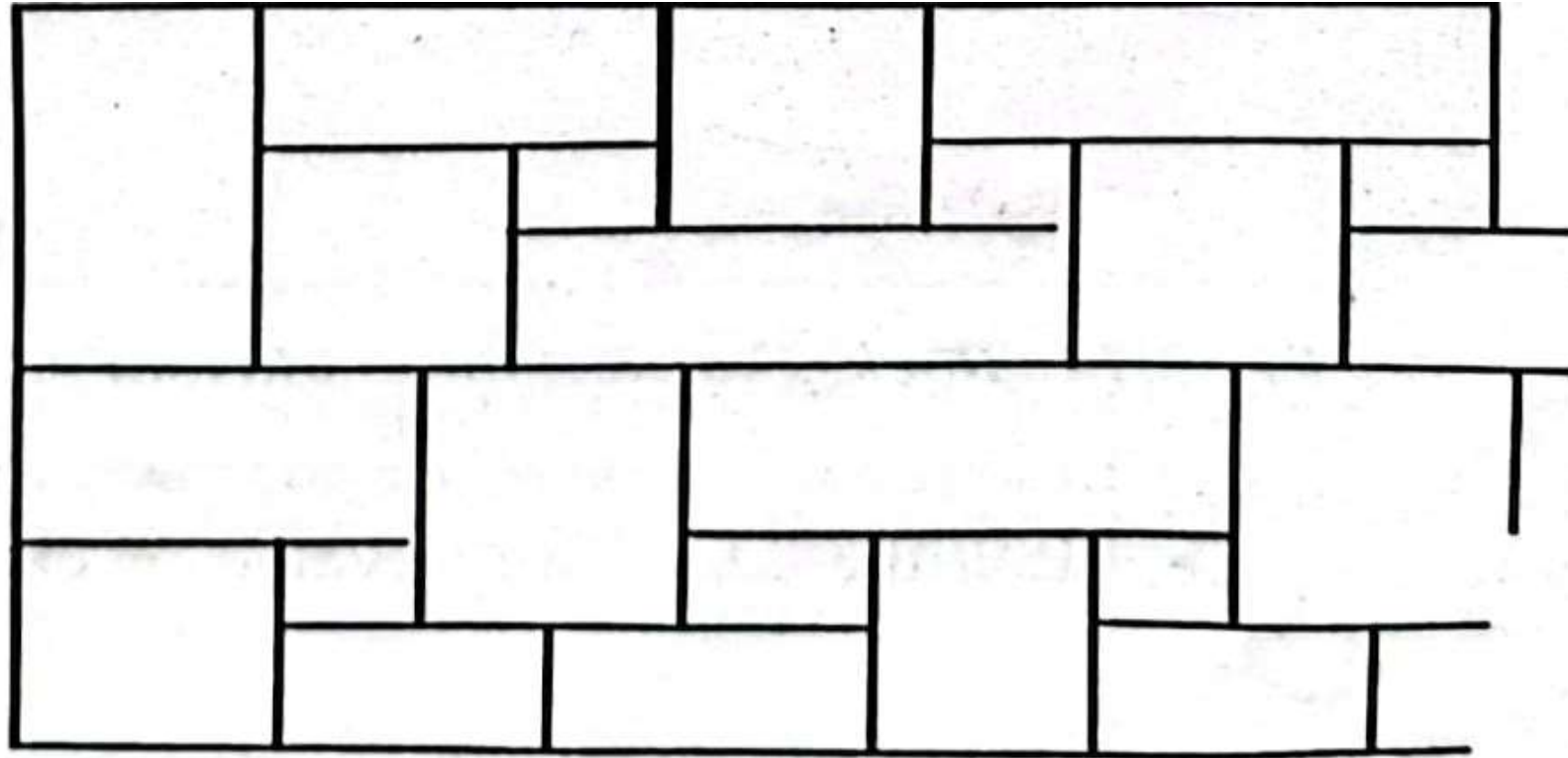
- In this type of stone masonry stone blocks of same height in each course are used.
- Every stone is fine tooled on all sides.
- It is an expensive type of stone masonry as it requires heavy labor and wastage of material while dressing.



Ashlar Fine or Coursed Masonry.

(ii) Random Coursed Ashlar Masonry

- This type of ashlar masonry consists of fine or coursed ashlar but the courses are of varying thicknesses, depending upon the character of the building.



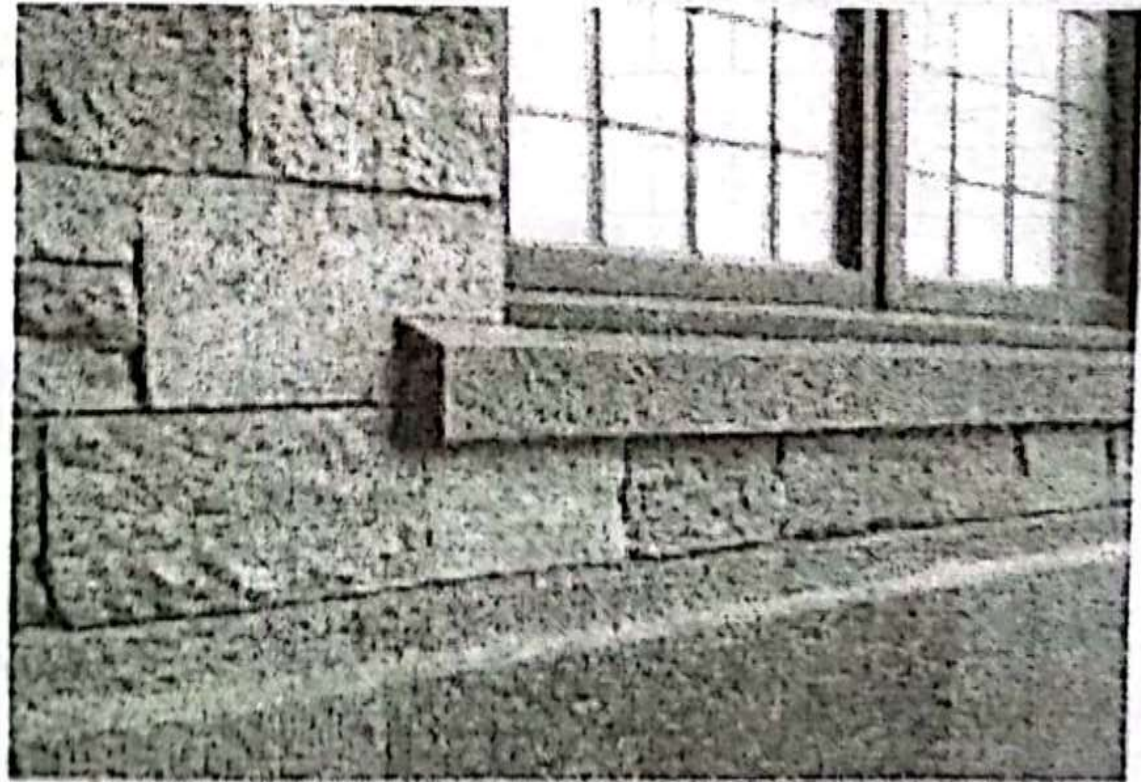
(b) Rough Tooled Ashlar Masonry

- In this type of ashlar masonry the sides of the stones are rough tooled and dressed with chisels.
- Thickness of joints is uniform, which does not exceed 6mm.



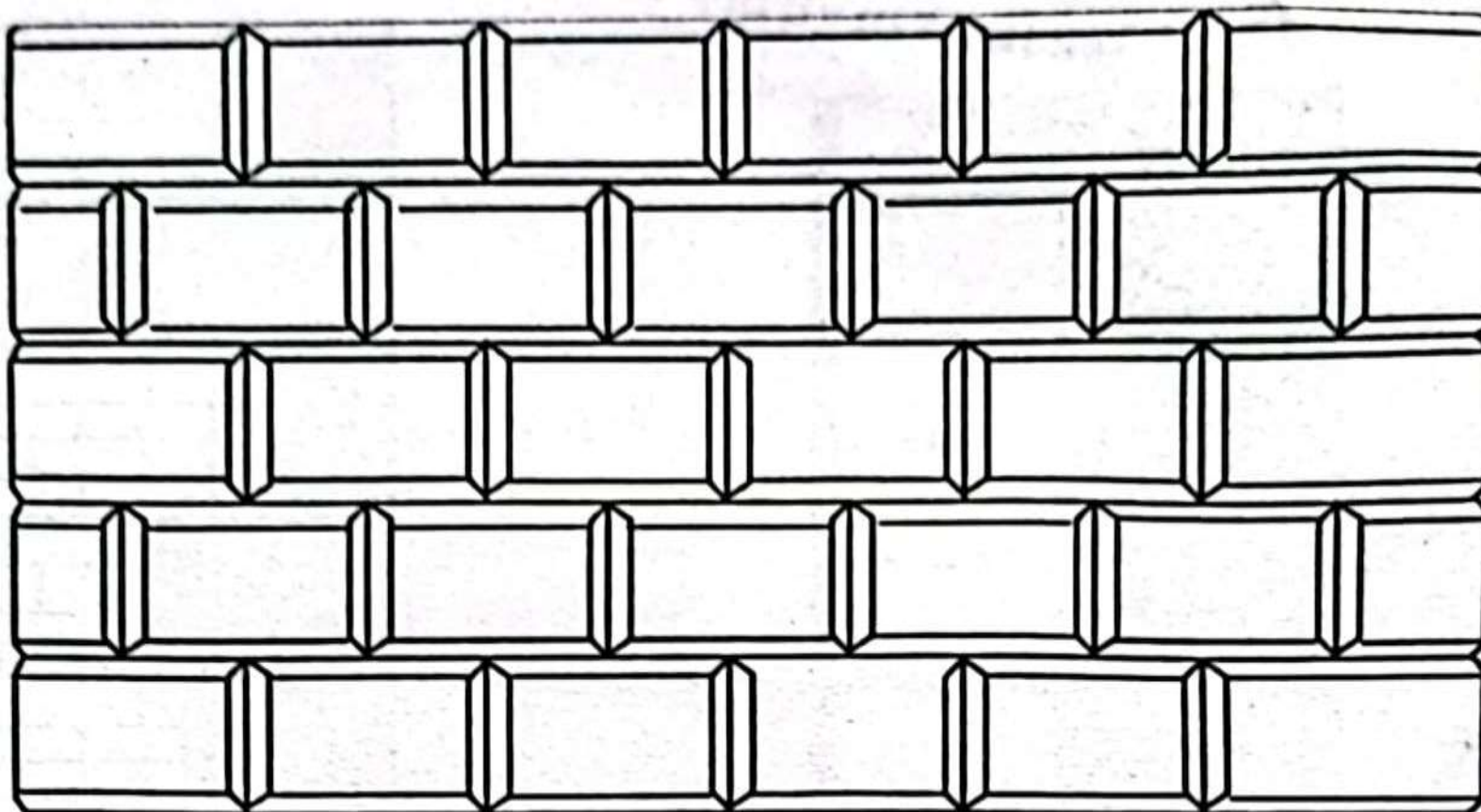
(c) Rock or Quarry Faced Ashlar Masonry

- This type of ashlar masonry is similar to rough tooled type except that there is chisel-drafted margin left rough on the face which is known as quarry faced.



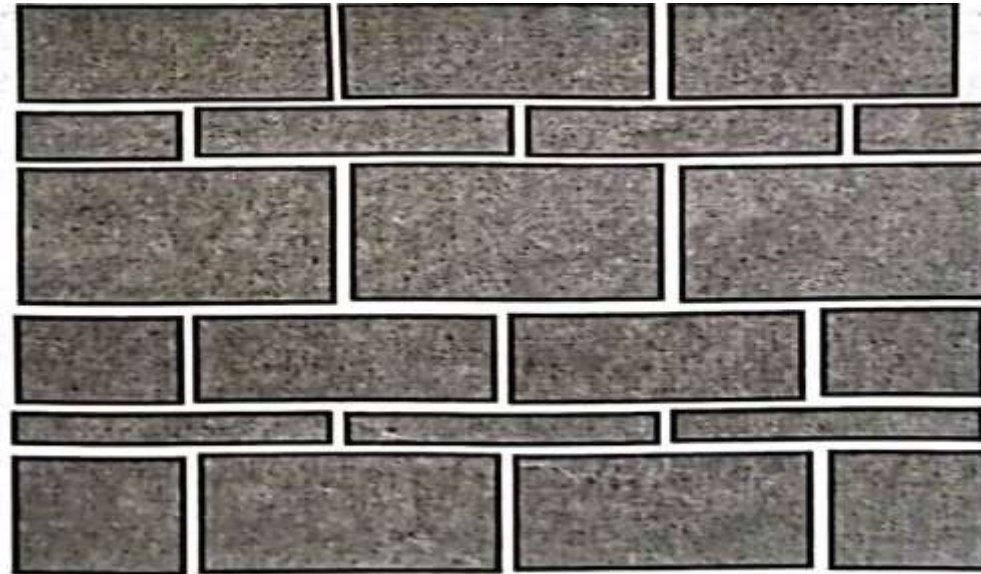
(d) Chamfered Ashlar Masonry

- It is similar to quarry faced except that the edges are beveled or chamfered to 45° for depth of 2.5 cm or more.



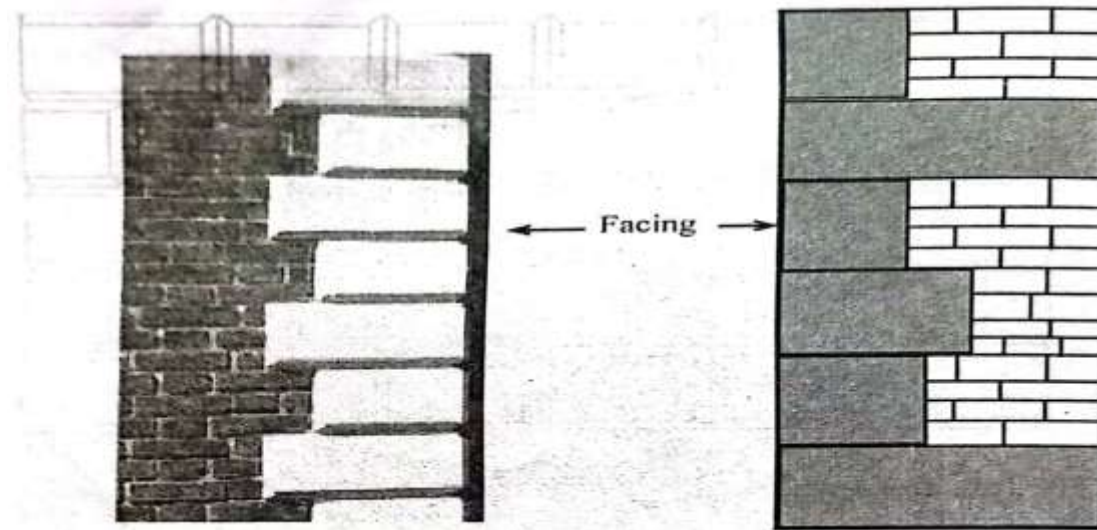
(e) Block in Course Ashlar Masonry

- It is the name given to a class of ashlar masonry which occupies an intermediate place between rubble and ashlar.
- The stones are all rec. and properly dressed.
- It resembles to coursed rubble masonry or rough tooled ashlar masonry.



(f) Ashlar Facing

- Is the best type of ashlar masonry.
- Since this type of masonry is very expensive.
- it is not commonly used throughout the whole thickness of the wall, except in works of great importance and strength.
- For economy the facing are built in ashlar and the rest in rubble.



Block Masonry; Autoclaved Aerated Concrete (AAC) Blocks & Solid Concrete Blocks:

Block Masonry:

- Construction technique using masonry blocks bonded with mortar.
- Used for walls, partitions, and load-bearing structures.
- Faster and more economical than traditional brick masonry.
- **Common Types**
 - AAC Blocks
 - Solid Concrete Blocks
 - Hollow Concrete Blocks

AAC (Autoclaved Aerated Concrete) Blocks:

Lightweight precast building material made from:

- Cement
- Lime
- Sand/Fly ash
- Water
- Aluminum powder

Uses

- High-rise buildings
- Partition walls
- Residential buildings



Advantages

- Reduces dead load of structure
- Faster construction
- Energy efficient
- Earthquake resistant due to low weight
- Less mortar consumption

Limitations

- Higher initial cost
- Brittle material
- Requires skilled labor
- Lower load-bearing capacity compared to concrete blocks

Solid Concrete Blocks

Dense concrete masonry units made from:

- Cement
- Sand
- Aggregate
- Water

Uses

- Load-bearing walls
- Retaining walls
- Foundations



- **Advantages**

- High structural strength
- Durable and long-lasting
- Low maintenance
- Resistant to weathering

Limitations

- Heavy weight increases dead load
- More labor required
- Poor thermal insulation
- Transportation cost is high

Load Bearing Structures and Framed Structures:

Introduction to Load Bearing Structure:

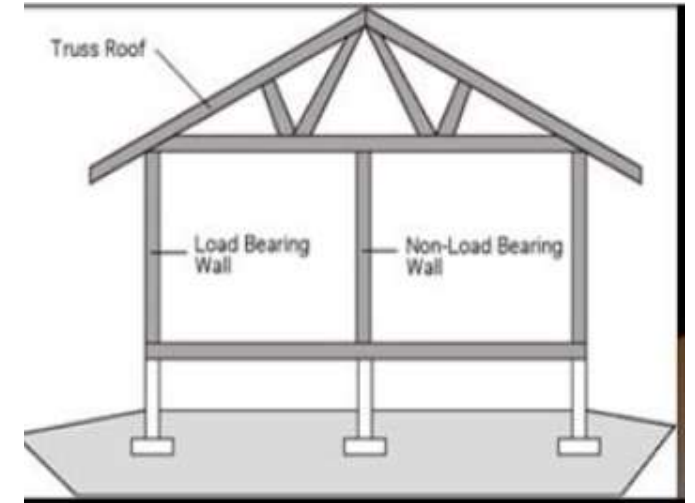
- A load bearing structure is a construction system where walls carry and transfer loads from roof and floors to the foundation.
- The walls support the entire building weight.
- Commonly used in small residential buildings.

Characteristics

- Thick masonry walls
- Limited height
- Simple construction
- Economical for low-rise buildings

Components of Load Bearing Structure:

- Foundation
- Load bearing walls
- Floor slab
- Roof
- Lintels and beams



Load Transfer Mechanism

- Roof load → Walls → Foundation → Soil
- Walls directly transfer structural loads.

Advantages

- Simple design
- Low construction cost
- Good thermal insulation

Disadvantages

- Thick walls reduce floor area
- Not suitable for high-rise buildings
- Limited flexibility in room arrangement

Applications of Load Bearing Structure:

- Residential houses
- Traditional buildings
- Small schools
- Rural constructions

Introduction to Framed Structure:

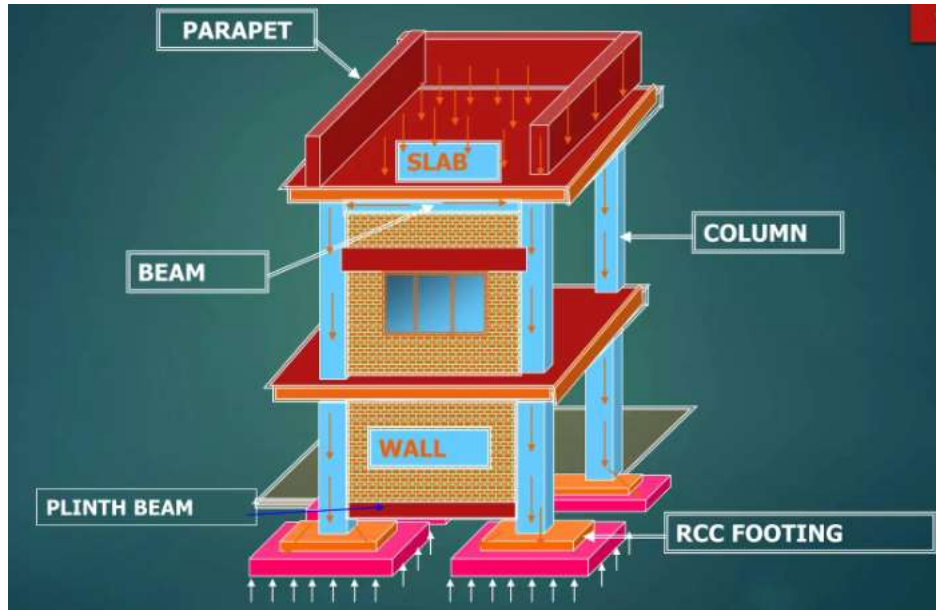
- A framed structure is a system where beams and columns carry loads to the foundation.
- Walls act only as partition walls.

Types

- Reinforced Cement Concrete (RCC) frame
- Steel frame structure

Components of Framed Structure

- Foundation
- Columns
- Beams
- Slabs
- Partition walls



Load Transfer in Framed Structure

- Slab load → Beam → Column → Foundation → Soil

Advantages

- Suitable for multi-storey buildings
- More flexible planning
- Larger openings possible
- Better earthquake resistance

Disadvantages

- Higher construction cost
- Requires skilled labor
- Complex construction process

Applications of Framed Structure

- High-rise buildings
- Commercial complexes
- Hospitals
- Office buildings
- Apartments

Reinforced Cement Concrete structure (Precast and cast-in-situ construction):

Introduction to RCC:

- Reinforced Cement Concrete (RCC) is a composite material made of:
 - Cement
 - Sand
 - Aggregate
 - Water
 - Steel reinforcement
- Concrete resists compression while steel resists tension.
- Widely used in buildings, bridges, dams, and other structures.

Advantages of RCC

- High strength and durability
- Fire resistance
- Long service life
- Economical maintenance
- Can be molded into different shapes
- Good resistance to weathering

Introduction to Precast Construction:

- Precast construction involves manufacturing RCC components in factories or casting yards.
- Components are transported to the site and assembled.

Examples

- Beams
- Columns
- Slabs
- Staircases
- Wall panels



Process of Precast Construction

- 1.Preparation of molds
- 2.Placement of reinforcement
- 3.Pouring concrete
- 4.Curing
- 5.Transportation to site
- 6.Erection and connection

Advantages

- Faster construction
- Better quality control
- Reduced site labor
- Less construction waste
- Durable and accurate components

Disadvantages

- Transportation difficulty
- Heavy lifting equipment required
- High initial setup cost

Applications of Precast Construction

- Bridges
- Flyovers
- Apartments
- Industrial buildings
- Boundary walls

Introduction to Cast In-Situ Construction

- In cast in-situ construction, concrete is poured and cured directly at the construction site.
- Formwork is used to shape the concrete.

Common Elements

- Foundations
- Beams
- Columns
- Slabs



Process of Cast In-Situ Construction

1. Formwork preparation
2. Reinforcement placement
3. Concrete mixing
4. Pouring concrete
5. Compaction
6. Curing
7. Removal of formwork

Advantages of Cast In-Situ Construction

- Flexible design
- Suitable for complex shapes
- No transportation problem
- Strong monolithic structure

Disadvantages

- Time-consuming
- Weather affects construction
- Requires more site labor
- Quality control may vary

Applications of Cast In-Situ Construction

- Residential buildings
- Water tanks
- Foundations
- Retaining walls
- High-rise buildings

Plastering Process and Defects in Plastering:

Introduction to Plastering

- Plastering is the process of covering rough wall or ceiling surfaces with mortar.
- It provides:
 - Smooth finish
 - Protection against weather
 - Decorative appearance
 - Improved durability

Common Materials

- Cement ,Sand, Lime, Water

Objectives of Plastering

- Protect masonry surfaces
- Improve appearance
- Increase durability
- Provide smooth surface for painting
- Resist moisture penetration

Plastering Process

1. Surface Preparation

Steps:

- Clean the wall surface
- Remove dust, grease, and loose particles
- Fill joints and holes
- Wet the surface before plastering

2. Preparation of Mortar

- Mortar is prepared by mixing:
 - Cement
 - Sand
 - Water

Common Mix Ratio

- 1:4 or 1:6 (cement:sand)

3. Application of Plaster

- **Procedure**

- Apply first coat (scratch coat)
- Level the surface
- Allow initial setting
- Apply finishing coat
- Smooth the surface with trowel

Thickness

- Usually 12 mm to 20 mm

4. Curing of Plaster

- Curing keeps plaster moist for proper strength development.

Methods

- Water sprinkling
- Wet covering

Duration

- Usually 7 days minimum

Defects in Plastering

- Defects occur due to:
 - Poor workmanship
 - Improper materials
 - Incorrect mixing
 - Inadequate curing

Cracks in Plaster

Causes:

- Shrinkage
- Poor curing
- Structural movement
- Excessive thickness

Prevention

- Proper curing
- Correct mortar mix
- Uniform thickness



Blistering and Flaking

Blistering:

- Formation of small swollen patches on plaster surface.

Flaking

- Peeling off thin layers from plaster surface.

Causes

- Poor bonding
- Excessive moisture
- Improper finishing



Efflorescence and Peeling

Efflorescence

- White salt deposits on plaster surface.

Causes

- Presence of soluble salts
- Moisture penetration

Peeling

- Separation of plaster from wall surface.

Prevention

- Use clean materials
- Proper waterproofing
- Adequate curing



Popping and Uneven Surface

Popping

- Formation of small holes due to expansion of particles.

Uneven Surface

- Irregular finish due to poor workmanship.

Prevention

- Use quality materials
- Proper mixing and finishing
- Skilled labor

Pointing:

- The technique of repairing mortar joints between bricks or other masonry elements.
- When aging mortar joints crack and disintegrate, the defective mortar is removed by hand or power tool and replaced with fresh mortar, preferably of the same composition as the original.
- It is the process of raking and applying rich mortar with desired finishing in the joints of masonry.



Types of pointing:

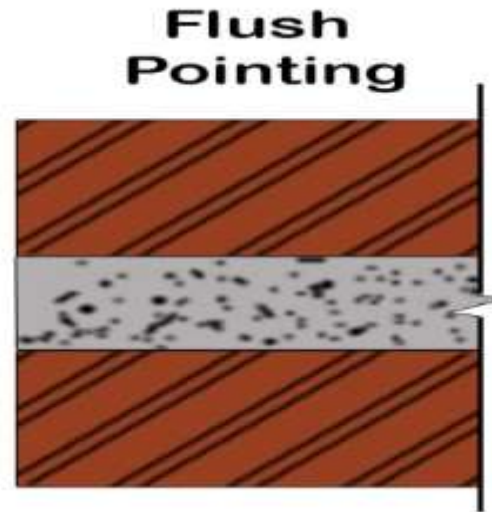
1) Beaded pointing

- This type of pointing is formed by a steel or iron rod with suitable size.
- Give good in appearance, But it can be easily damaged.



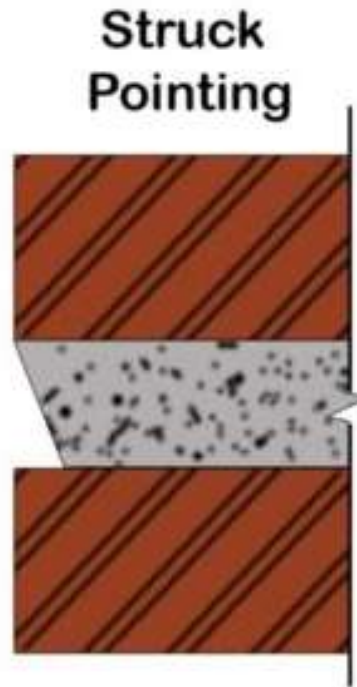
2) Flush pointing

- This type of pointing is formed by removing the excess mortar from the joint.
- The joint is made flush with the face.
- This type of joint does not give good appearance.
- But it is durable as it does not provide any space for collection of dust, water, etc.



3) struck/ cut pointing

- The face of pointing is kept inclined.
- The upper edge of joint is about 10 mm inside the face of masonry.
- It will not form a satisfactory joint as water will be collected in the joint.



4) V-grooved or bucket handle pointing

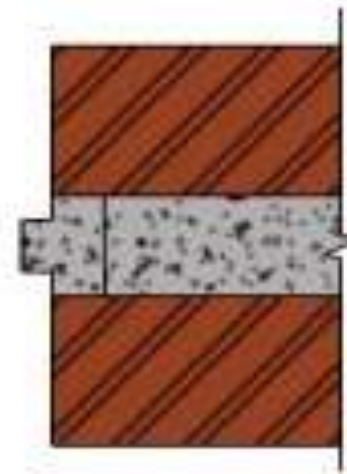
- In this type of pointing, a V-shaped groove is formed in the mortar joint.
- Give attractive appearance to the surface.
- Mostly used for superior type work.



V-grooved
Pointing

5) Tuck pointing

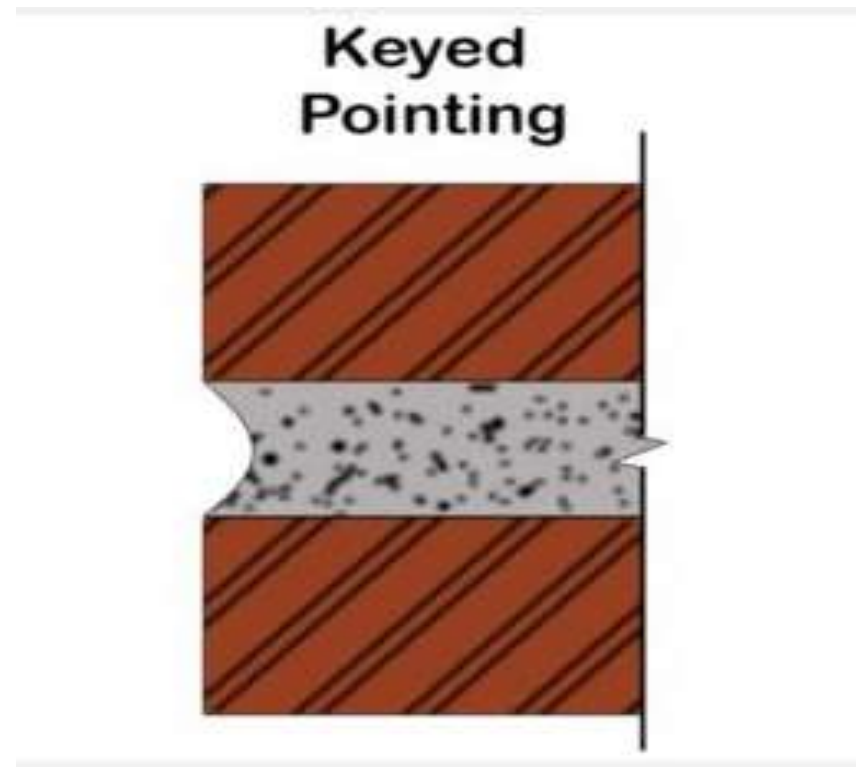
- In this type of pointing, a groove is formed at the center of joint.
- The width and depth of groove are 5 mm and 3 mm respectively.
- It is filled with white cement or putty.



**Tuck
Pointing**

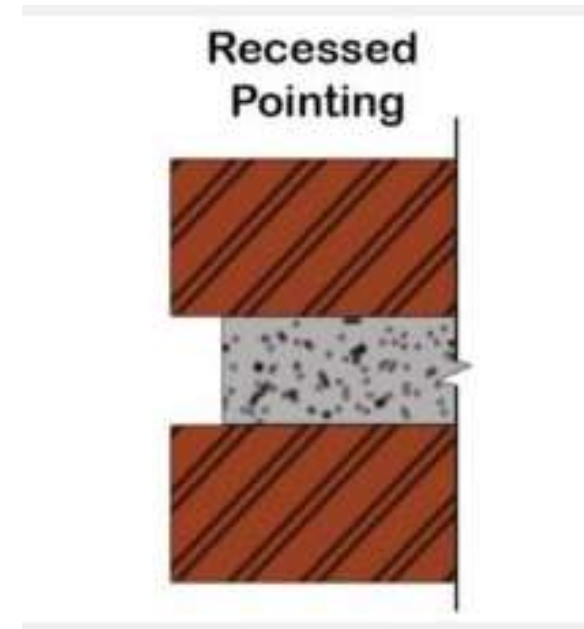
6) Rubbed/ keyed/ grooved pointing

- In this type of pointing, a groove is formed by a pointer.
- This type of pointing gives better appearance and is generally adopted.



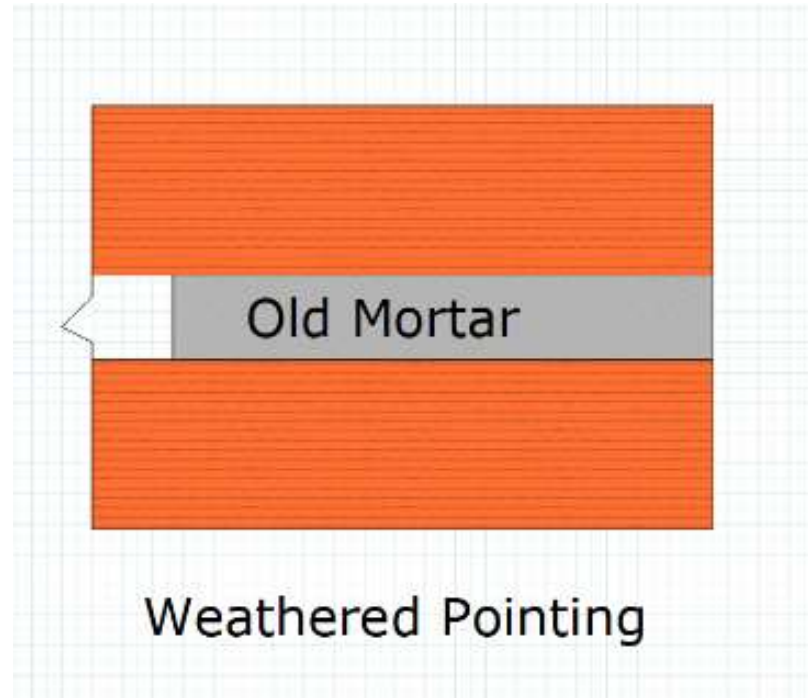
7) Recessed pointing

- In this type of pointing the face of the pointing is kept vertical and it is pressed inside the wall surface by a suitable tool.
- Mortar is pressed inside of joint by 5mm or more by suitable tool.
- Face of pointing is kept vertical.



8) Weathered (V) pointing

- V- shaped projection of mortar in joints is made.



Painting works in wooden, Metal and Masonry Surface:

Paint:

- Paint is the liquid material applied on timber, metal or masonry in order to protect or decorate them.
- It is composed of two main components: pigment and organic binders.

Function of paints:

- It provides a smooth and attractive surface.
- It prevents corrosion in metal structures.
- It is used to give a decorative effect on the surface.
- It prevents decay of wood-work.

Characteristic of Ideal Paint:

- It should have a good spreading power.
- It should work smoothly and freely and be capable of being laid in a thin coat with the brush.
- It should form durable, tough and resistant to wear film on drying.
- Color of paint should not fade or change.
- It should not crack on drying.
- It should give a smooth and pleasing appearance showing no brush marks on drying.
- It should not damage the painted surface.

Painting on wood work:

a) Painting process on new wood work

i) Preparation of surface

ii) Knotting

Patent knotting

Size knotting

Lime knotting

iii) Priming

iv) First coat

v) Finishing coat

b) Painting process on old wood work

- i) Preparation of surface
- ii) Coating of paint

Painting on metal:

- Remove the old paint. Before you start, make sure you put down a drop sheet and wear a dust mask and gloves.
- Spray paint the metal with primer. After you've removed all of the flaky paint, you're ready to paint.
- Use a brush to paint your metal.
- Apply the second coat

Painting works on masonry:

Procedure of painting on masonry works:

- Clean the surface
- Make any repair if necessary
- Seal and prime
- Paint the surface

Introduction to Steel Construction:

- Steel construction uses steel as the primary structural material.
- Commonly used in buildings, bridges, towers, and industrial structures.
- Provides high strength and flexibility in design.

Advantages:

- High strength-to-weight ratio
- Fast construction and easy fabrication
- Durable and long-lasting material
- Easy to modify and expand
- Recyclable and environmentally friendly

Disadvantages:

- Chance to corrosion if not protected
- Strength decreases at high temperatures
- Requires skilled labor and equipment
- Higher initial cost
- Regular maintenance may be needed

Bolted Connections:

- Uses bolts and nuts to connect steel members
- Easy to assemble and dismantle
- Suitable for site construction
- Requires drilled holes in members

Advantages of Bolted Connections:

- Quick installation process
- Less skilled labor required
- Easy inspection and replacement
- No heating required during connection

Disadvantages of Bolted Connections:

- May loosen under vibration
- Requires accurate hole alignment
- Connections may appear bulky

Welded Connections:

- Steel members are joined by melting and fusing
- Provides rigid and strong connections
- Commonly used in factories and heavy structures

Advantages of Welded Connections:

- Strong and rigid joints
- Neat appearance without bolts
- Efficient stress distribution
- No need for drilled holes

Disadvantages of Welded Connections:

- Requires skilled workers
- Difficult to inspect internally
- Heat may affect material properties
- Repair and dismantling are difficult

Introduction to Formwork:

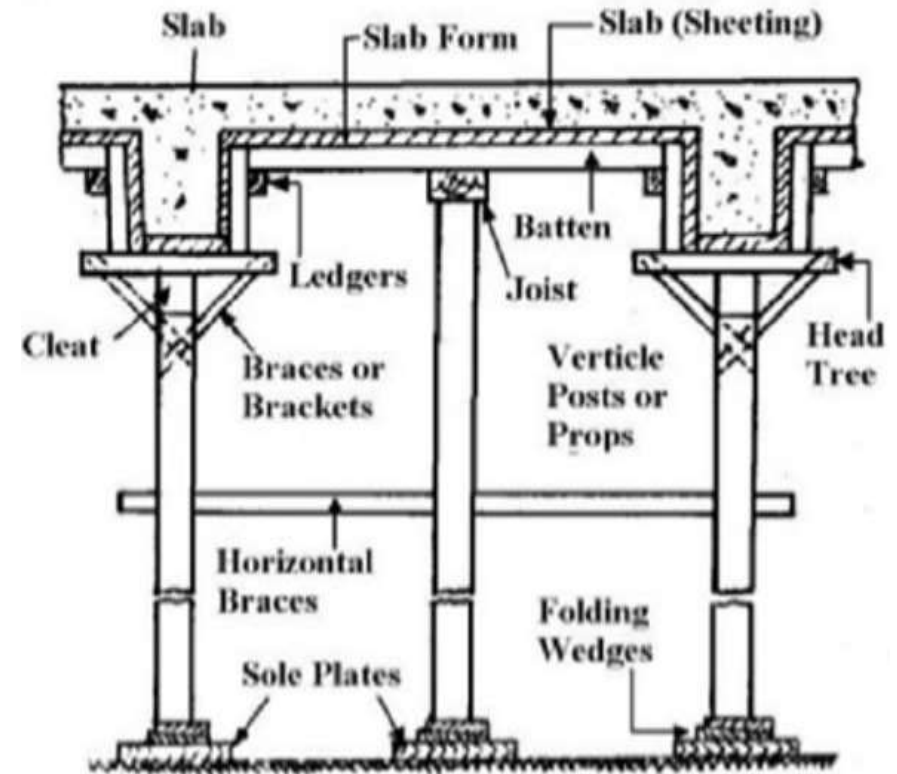
- Formwork is a temporary structure used to support fresh concrete.
- It gives shape and support until concrete gains sufficient strength.
- Common materials: timber, plywood, steel, aluminum, and plastic.

Requirements of Good Formwork:

- Strong enough to carry loads
- Rigid and properly aligned
- Easy to remove without damaging concrete
- Leak-proof joints
- Economical and reusable

Slab Formwork:

- Used for casting RCC slabs.
- Consists of sheeting, joists, bearers, and props.
- Supports wet concrete until hardening.



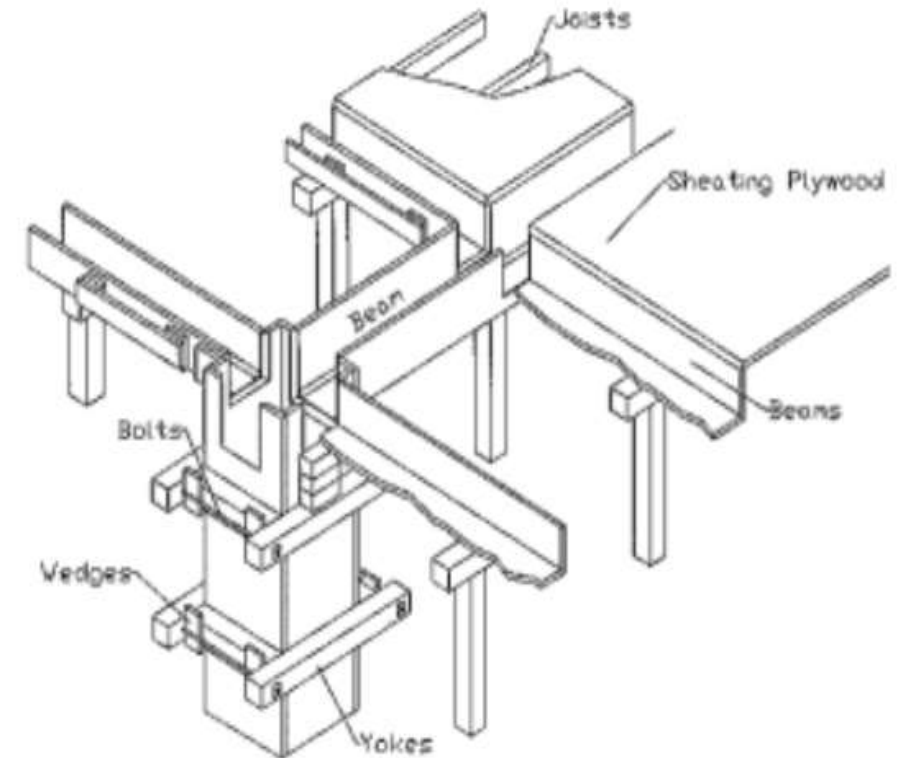
Formwork For Beams And Slabs

Components of Slab Formwork:

- Plywood or sheeting
- Joists and bearers
- Vertical props
- Bracing for stability

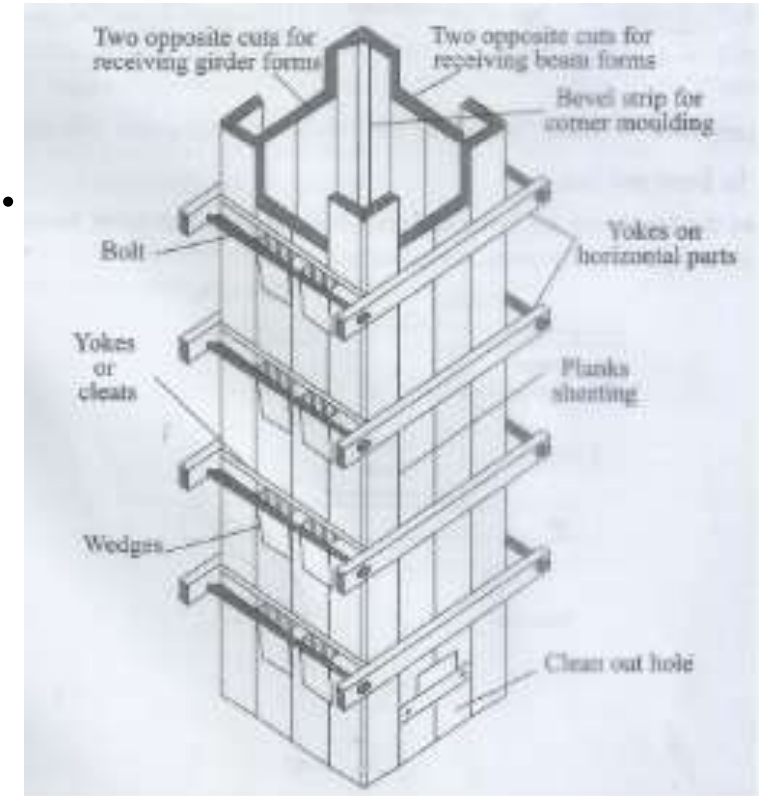
Beam Formwork:

- Used for casting RCC beams.
- Includes bottom soffit and side panels.
- Supported by props and bracing.



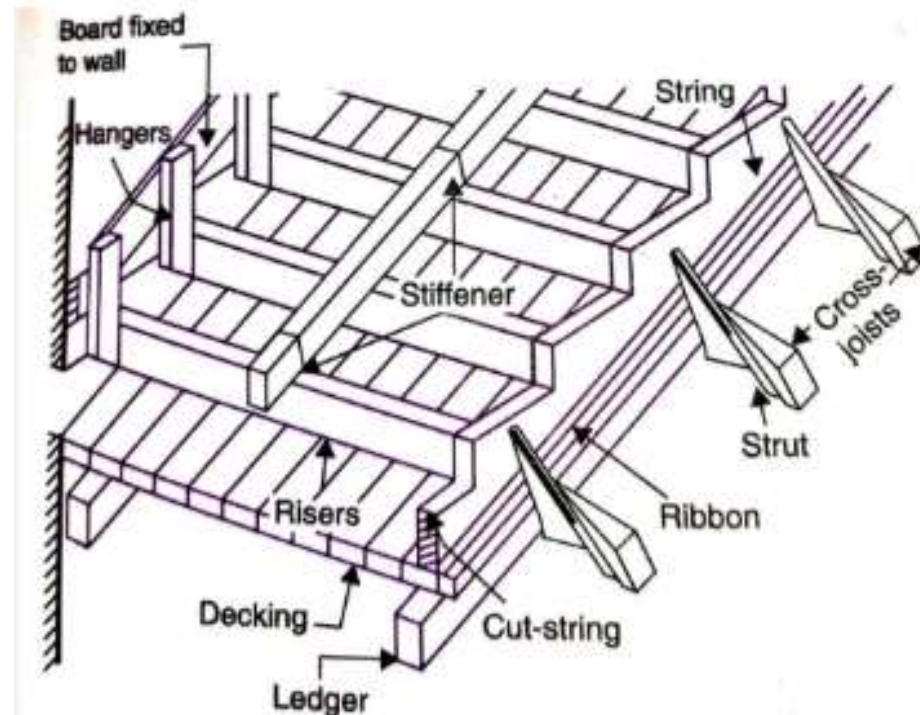
Column Formwork:

- Used to cast RCC columns.
- Usually consists of vertical panels and yokes.
- Must resist pressure of fresh concrete.



Staircase Formwork:

- Used for casting RCC staircases.
- Supports waist slab, risers, and treads.
- Requires careful measurement and alignment.



Thank You